Open Knowledge for a Sustainable Future: Research, Ethics, and Wikipedia

Week 5 — FAIR: making data findable and reusable

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Today

1. FAIR Data: Findable, Accessible, Interoperable, Reusable
2. Open Data Gets Better
3. Open Licenses and Publications
4. Practice!

Roadmap

1. FAIR Data: Findable, Accessible, Interoperable, Reusable
   - Findable
   - Accessible
   - Interoperable
   - Reusable
2. Open Data Gets Better
3. Open Licenses and Publications
4. Practice!

Why FAIR Data?

- **Findable**: other researchers (and future you) can discover your data
  - If no one can find it, it might as well not exist
- **Accessible**: once found, the data can be retrieved
  - Even years later, with stated rules if access is restricted
- **Interoperable**: data from different projects can be combined or compared
  - Shared formats and vocabularies, for computers and for people
- **Reusable**: data can be used beyond its original purpose
  - Documentation and a licence let others build on your work

Sources: (Wilkinson et al., 2016), FAIR Data 101 (ARDC, CC BY 4.0; accessed 2026-09-20)

Findable: why it matters

- Data no one can find has no value
- Titles, metadata and persistent identifiers (DOIs) make your work visible to people and to search engines
- Data that is easy to find is easy to cite: you get the credit

Making data findable

- Use descriptive, consistent names for datasets and files
- Add rich metadata: who, what, when, where, why
  - Use standard terms, so people can find them easily
- Register your dataset in a trusted repository with a permanent identifier (e.g., Zenodo, Dataverse)
- Cross-link: paper $\leftrightarrow$ dataset $\leftrightarrow$ code
- This applies to governments and companies as well as academia
  - Making data findable should be part of making the data!

Identifiers for works: DOI

- We need persistent identifiers: an ISBN names a book wherever it is shelved; a **Digital Object Identifier** names a paper, dataset or program wherever it is hosted
- `10.1007/s10579-008-9062-z`: a prefix for the publisher (10.1007 is Springer), a suffix they choose
- https://doi.org/10.1007/s10579-008-9062-z redirects to wherever the paper is now
  - In 2009 the record pointed to `springerlink.com/content/v7q114033401th5u/`: that URL is dead
  - Today it goes to `link.springer.com`; the DOI did not change
- Issued by registration agencies: Crossref for articles and books, DataCite for data and software (Zenodo)
  - About 275 million assigned so far
  - Publishers pay for theirs; Zenodo gives you one free

Source: What is a DOI? and Key facts (DOI Foundation; accessed 2026-10-27)

What a DOI knows

- The DOI carries its metadata: authors, title, venue, year, pages, current URL
- So you can ask it for the citation:

```
$ curl -L -H "Accept: text/x-bibliography; style=apa" https://doi.org/10.1007/s10579-008-9062-z
Bond, F., Fujita, S., & Tanaka, T. (2008). The Hinoki syntactic and semantic treebank
of Japanese. Language Resources and Evaluation, 42(2), 243-251.
https://doi.org/10.1007/s10579-008-9062-z
```

  - This is what your reference manager does (week 4)
- Citations can be counted: one identifier, not ten URLs for the same paper
- No DOI (older papers, LingBuzz, websites)? Give a stable URL and the date you read it
- A DOI that does not resolve is a warning sign (week 6)

Identifiers for people: ORCID

- A DOI identifies a work; an ORCID iD identifies the person who made it
  - 16 digits: Francis's is 0000-0003-4973-8068
  - Free, run by a non-profit; it stays with you when you change institution or name
- It solves the name problem from week 4
  - Nurril Hirfana binte Mohamed Noor, Nurril Hirfana, or N. H. Mohamed Noor?
  - Or the many F. Bonds
- Journals, funders and repositories ask for it when you submit, and put it in the metadata (as in the `.zenodo.json` two slides on)
- Wikidata links people to it too: property P496
- Registration at orcid.org is free

Metadata inside the file: this PDF

`pdfinfo -custom wk-05-fair.pdf` (or *File* → *Properties* in a PDF reader):

```
Title:      Open Knowledge for a Sustainable Future: Research, Ethics, and Wikipedia - Week 5 - FAIR:
↪  making data findable and reusable
Author:     Francis Bond (Academic), Pavel Bednařík (Wiki)
Subject:    Open Knowledge for a Sustainable Future: Research, Ethics, and Wikipedia (Palacký
↪  University Olomouc)
Keywords:   open knowledge, research ethics, Wikipedia, FAIR data, CARE principles, academic writing,
↪  peer review
License:    CC BY 4.0
LicenseURL: https://creativecommons.org/licenses/by/4.0/
Producer:   LuaTeX-1.22.0
```

- A few lines in the LaTeX preamble; in Word: *File* → *Info* → *Properties*
- Search engines and reference managers read these fields
- The metadata travels with the file: email it, and the licence goes too

Metadata for a repository: `.zenodo.json`

```
{
  "title": "Open Knowledge for a Sustainable Future: Research, Ethics, and Wikipedia",
  "description": "Course materials (slides, handouts and website) for a course at Palacký University
  ↪  Olomouc on writing (both academic and for Wikipedia), finding, judging and citing sources, FAIR
  ↪  and CARE data, feedback and peer review.",
  "upload_type": "lesson",
  "creators": [
    {"name": "Bond, Francis", "affiliation": "Palacký University Olomouc",
     "orcid": "0000-0003-4973-8068"},
    {"name": "Bednařík, Pavel", "affiliation": "Wikimedia ČR"}
  ],
  "license": "cc-by-4.0",
  "language": "eng",
  "keywords": ["open knowledge", "research ethics", "Wikipedia", "FAIR data",
               "CARE principles", "academic writing", "peer review"],
  "related_identifiers": [
    {"identifier": "https://bond-lab.github.io/Open-Knowledge/",
     "relation": "isDocumentedBy", "resource_type": "other"}
  ]
}
```

Put this in a GitHub repository connected to Zenodo: each release is archived, with a DOI.

Accessible: why it matters

- Well-described data is useless if no one can reach it
- Others can download or view your work, now and in the future
- The access rules say who can use the data, and under what conditions
- Accessible does not mean open: sensitive language or cultural data can have access rules, as long as they are stated (more next week)

Making data accessible

- Store your data in repositories that use open web standards (HTTP/HTTPS)
  - Use persistent repositories
  - Support persistent repositories
  - Company data is perishable: when the product goes, so does the data
- Keep metadata available even if files must be restricted
- Explain how others can request access if necessary
- Test from a new computer: can someone else follow your instructions?

Interoperable: why it matters

- Most questions need data from more than one source
- Your data can connect with others', across languages, disciplines and tools
- Shared standards stop data ending up in silos only one group can use

Making data interoperable

- Use open, non-proprietary formats (CSV, JSON, XML)
  - Text is better than binary, as you can always read it
- Follow community conventions for terminology and encoding
- Provide clear documentation of structure and meaning
- Link to external vocabularies or identifiers when possible
  - ISO 639 for languages and language groups
    - English: en, eng (ISO 639-1, 2&3)
    - Czech: cs, cze/ces, ces (ISO 639-1, 2-B/T, 3)
    - Since ISO 639:2023 the parts are called sets; the codes are unchanged

Reusable: why it matters

- Good data should outlive the project that created it
- Reuse lets others verify results, test new questions, and save time
- Documented, licensed data saves others from collecting it again
- Many agencies produce data as their job; if no one uses it, the effort is wasted
- Reuse is the payoff for the rest of FAIR

Making data reusable

- Include full context: how the data was collected, cleaned, and analysed
- Choose a clear licence (e.g., CC BY or CC BY-SA)
- Describe known limitations or ethical constraints
- Provide examples of how to cite or acknowledge your dataset

How the Open Multilingual Wordnet asks to be cited (omwn.org)

"If you use these wordnets, please cite the original projects who created them, if you got value from this aggregation/normalization, please cite Bond and Paik (2012)."

References

Francis Bond and Kyonghee Paik (2012) A survey of wordnets and their licenses In *Proceedings of the 6th Global WordNet Conference (GWC 2012)*. Matsue. 64–71

Roadmap

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Many Eyes Make Bugs Shallow

- Linus's Law, as Raymond (1999) named it:
  - "Given enough eyeballs, all bugs are shallow"
- In open source software every user is a tester, and someone will find the fix obvious
  - The person who finds a bug is usually not the one who fixes it
- The same holds for data
  - A closed dataset is checked by the people who made it
  - An open dataset is checked by everyone who uses it
  - Each new use looks at the data from a new angle, and finds new errors
- From our own work: when we gathered the world's wordnets in 2008, Princeton WordNet had a loop in its hierarchy and the Global WordNet grid had broken encodings
  - No one had noticed, because few people had tried to combine them

Many Small Contributions

- Each contribution is small: a typo, a wrong language code, a missing entry, a dead link
- The sum is a resource no one group could build
  - Wikipedia, OpenStreetMap, Wikidata, the wordnets, Linux
- Finding the error is the hard part; the fix is usually trivial
  - But only if the finder can report it, or fix it themselves
- Fixes compound: a dataset that has been used for ten years has had ten years of corrections
- Credit matters: people contribute more when their fixes are seen and acknowledged
  - A contributors list, a changelog entry, a line in the paper

Closing the Loop: Make It Easy to Contribute

- Give people somewhere to report problems: an issue tracker (GitHub, GitLab) or a contact address in the README
- Version the data, so a fix is a recorded change, not a new file
  - Each release gets a DOI (as in the `.zenodo.json`): papers can cite the version they used
- Say how to contribute: a CONTRIBUTING file, a changelog, and validators that check the data
- Act on reports, and say thank you: an acknowledged fix brings the next one
- A licence that forbids redistribution breaks the loop: the fix cannot be shared
  - Which brings us to licences
- These slides are on GitHub under CC BY: if you find a mistake, tell us!

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The Open Definition (Open Knowledge Foundation)

- **Open data**: free access plus legal permission to use, modify and share, for anyone
- The Open Definition 2.1 puts it in one sentence:

> "Knowledge is open if anyone is free to
> **access**, **use**, **modify**, and **share** it
> — subject, at most, to measures that preserve provenance and openness."

- Provenance: say where it came from (attribution); openness: keep it open (share-alike)

Open licences

- For text, data and media:
  - **CC BY**: reuse with attribution (these slides)
  - **CC BY-SA**: attribution and the same licence (share-alike): Wikipedia
  - **CC0** / public domain: no restrictions: Wikidata
  - **ODC-BY**: for databases; attribution only
  - **ODbL**: for databases; attribution and share-alike: OpenStreetMap
- For code:
  - **MIT** (also BSD, Apache 2.0): attribution only; the commonest on GitHub
  - **GPL**: share-alike (copyleft): Linux, R
- Not open: **CC BY-NC** (no commercial use) and **CC BY-ND** (no changes) fail the Definition, however common they are
- No licence means all rights reserved: no one can legally reuse it, however open you meant it to be

When not to use open licences

- Some data should not be released openly, because openness could cause harm
- Examples:
  - Personal or medical information that identifies individuals
  - Cultural or linguistic materials shared under community protocols
  - Locations of endangered species or sacred sites
  - Data collected without full, informed consent for public reuse
- Use restricted or tiered access, or licences that reflect community agreements
- Next week: the CARE principles for exactly these cases

Where to publish: routes to open access

- A licence says what readers are allowed to do. The **route** decides whether they can read the work at all
- **Gold**: the journal publishes the final version openly, and charges the author an article processing charge (APC): hundreds to thousands of euros, usually paid by a funder or institution
- **Diamond**: open to readers and free to authors, funded by universities or scholarly societies. Much of linguistics works this way
- **Green**: publish anywhere, then deposit the accepted manuscript in a repository — your university's, or a subject archive such as arXiv or LingBuzz
- **Predatory venues** take the APC and skip the peer review. Check the journal against the DOAJ, and see whether you recognise anyone on the editorial board

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Due today: your annotated bibliography

- **Five or more sources**, in APA 7, with **two or three sentences** on each
- Each annotation answers three questions:
  - What does this source *say* that bears on your question?
  - How far can you *trust* it — who wrote it, where did it appear, what is the evidence behind it?
  - What will you *use* it for: background, evidence, a position you argue against?
- It is not a summary exercise. An annotation that could have been written without reading the source is worth nothing to you in December, when you are writing the paper and cannot remember which one said what
- Not graded on its own — it is the scaffolding for the paper, and we use it today

Auditing sources for FAIR and CARE issues

- **30 minutes, in pairs.** Work from the annotated bibliography you brought today
- **Part 1 (10 min): pitch your topic.** Tell your partner the question you want your paper to answer, then read them two of your annotations: your best source and your shakiest. Your partner asks: how do you know that one is any good?
- **Part 2 (15 min): audit one source each.** Pick the shaky one and check it:
  - **FAIR:** can you find it by DOI or stable URL, can you read it, is its data or method reusable, is the licence clear?
  - **CARE:** whose data or words does it rest on? Did they consent? Who benefits from it being published? (more next week)
- **Part 3 (5 min):** decide together — keep it, replace it, or cite it with its limitations stated. Rewrite that annotation before you leave

Your first-draft paper: fitting the scientific style

- Next assignment: the first draft of a short academic paper
- Use the conventions of scientific writing:
  - Clear research question and motivation
  - Evidence-based argument supported by credible sources
  - Neutral tone; cautious interpretation; accurate citations
- Both pieces of work you have done so far help:
  - The three-paragraph comparison of texts on water use in data centres → weighing sources against each other, and citing them in APA
  - Wikipedia → writing neutrally and clearly for others
- What I want to see: how well you use your sources to answer the question, as accurately as possible
- In the peer-review session you will read and comment on each other's papers

Acknowledgements I

- **FAIR material** draws on FAIR Data 101 (Australian Research Data Commons, CC BY 4.0) and Wilkinson et al. (2016).
- **AI disclosure: GPT-4o** in ChatGPT (OpenAI, 2025) was used to format the references and to draft an early version of these slides. **Claude Fable 5.1** (Anthropic, 2026a) edited the wording of the deck against Francis's style guide and drafted the section *Open Data Gets Better* from his outline.
- **Claude Opus 5** (Anthropic, 2026b) was used to check the slides: to resolve every DOI against Crossref, fetch every URL, and compare each quotation with its original. The prompt was of the form

```
Verify every claim, DOI, URL and quotation against a real source. Where you cannot
verify something, say so rather than guessing --- never invent a reference.
```

  Each correction names the source it was checked against. Responsibility for what remains is mine.

References I

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